

Lab 12: Baseball

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Part 1

1. What are some questions we can answer about baseball (or other sports) using statistics?

We can analyse the player performance using statistics in sports. For example, in ultimate, we can record the number of passes passed before the scoring of goal in end-zone and the total number of wins to analyse (find a correlation) the chance of wins of the player.

2. How do you think publicly available baseball data is collected? What sources of error?

The publicly available baseball data could be human and sensor collected from each of the public big games played by MLB, and recorded into publicly available database such as the Lahman database. The sources of error could be human error of the variation in clicking the button to count the number of strikes. There could also be sensor error, such as the minimal value that a sensor is able to record (i.e a sensor may not be able to differentiate 90.4 and 90.5)

3. What is the unit of observation for the Teams data set? What about for the Batting data set?

The observation unit of the Teams data set is one team. The observation unit of the Batting data is one player.

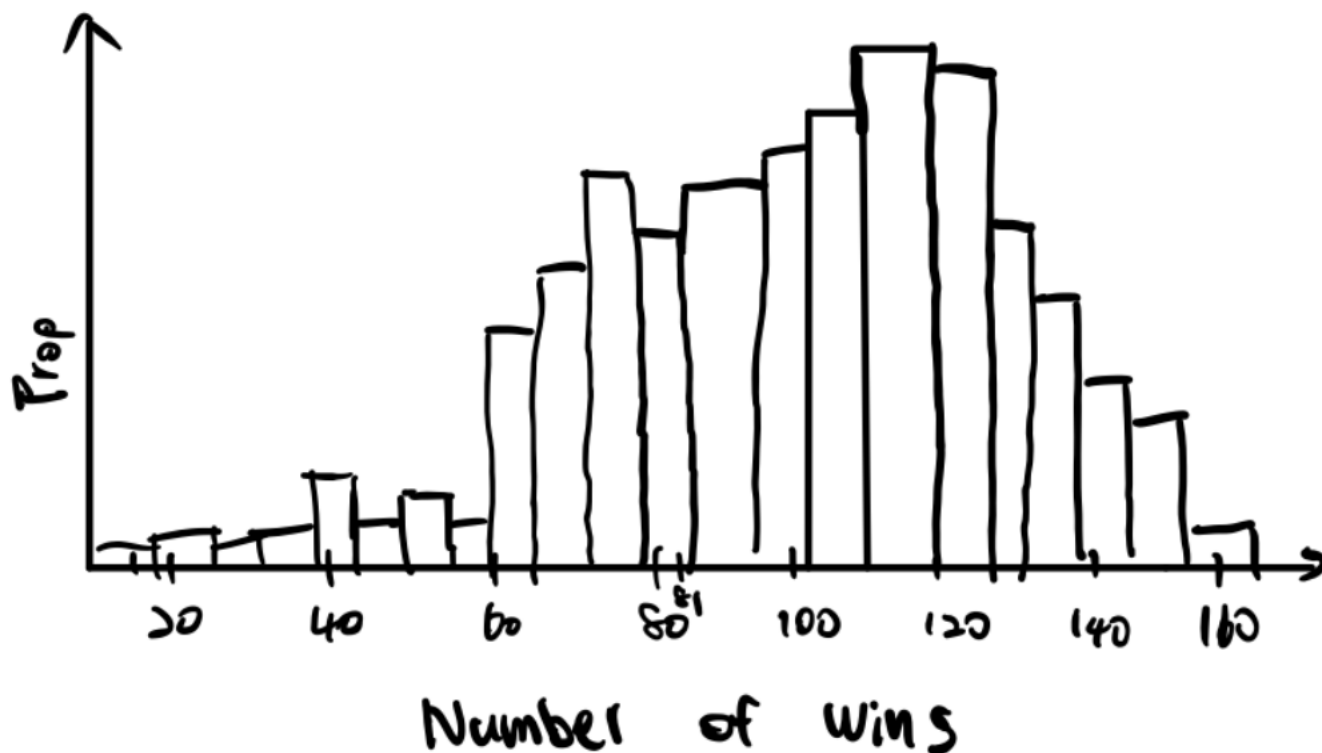
4. What is a question you could answer using the Teams dataset but not the Batting data set, and vice versa?

You can answer the question of “What’s the proportion of the Runs the Team scored compared to the Runs Opponent scored?” for Teams dataset but not the Batting data set. You can answer the question of “How many times did the batters performed strikeout, hits, and walks?” from the Batting data set but not the Teams

5. What is a question that we would need more granular data than the Teams and Batting dataset provide to answer?

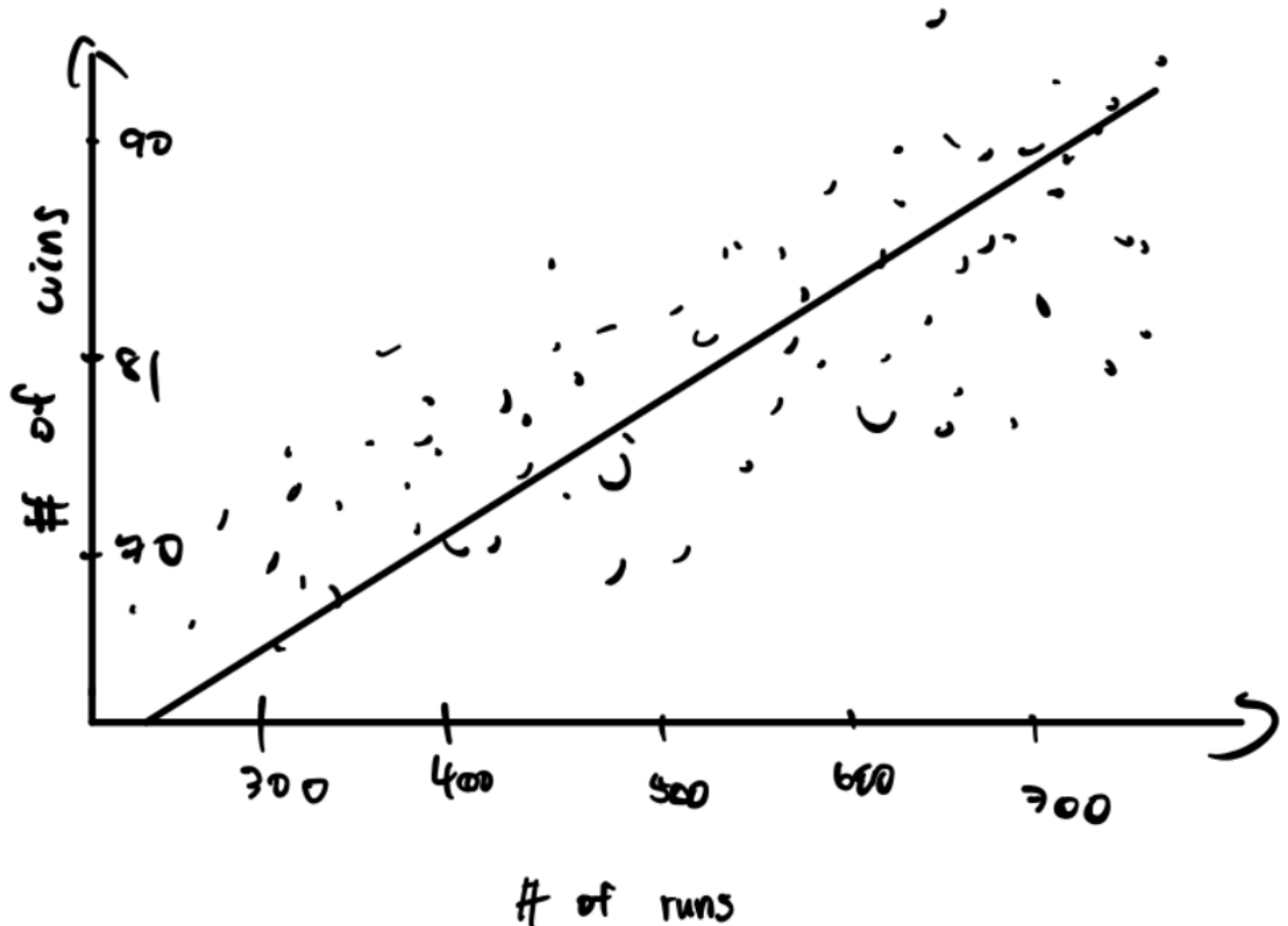
To answer “What is the speed of the baseball throwed by the pitcher?” would need more granular data than the Teams and Batting data set.

6. Roughly since 1962 MLB teams have played 162 games in a season. What do you think the distribution of wins looks like? Sketch a plot and describe.



I speculate that the MLB is good, so they would have more win then loss. The average is 81 if the proportion of win and loss are 50%. Thus, the distribution could be bell-shaped and left-skewed with mode around 85. There would be variation in the performance of MLB throughout the years, so it could be multimodal. The graph that best fit is the histogram.

7. What do you think the relationship between wins and runs looks like? Sketch a plot and describe



The relationship between win and number of runs could be strong, positive, and linear. As the number of win increase, the chance of winning also increases. There could be no conspicuous presense of outlier

8. Some people believe analytics is ruining baseball because teams are more cautious which makes the games less entertaining. Do you agree or disagree? Why?

I disagree, at the end of the day, the team wants to maximize their probability to win. The analytics could serve as something that is high risk, high reward. It could help the team to better visualize their ability compared to other teams and find specific areas to improve by looking at the different variable recorded.

Part 2

```
library(tidyverse)
```

```
## -- Attaching packages ----- tidyverse 1.3.1 --
```

```
## v ggplot2 3.3.5    v purrr   0.3.4
## v tibble  3.1.6    v dplyr   1.0.8
## v tidyr   1.2.0    v stringr 1.4.0
```

```
## v readr 2.1.2 v forcats 0.5.1

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()

library(infer)
library(Lahman)
data(Teams)
```

9. Subset the Teams dataset to only include years from 2000 to present day. What are the dimensions of this filtered dataset?

```
Teams %>%
  filter (yearID >= "2000", G>60) %>%
  head(10)
```

```
##   yearID lgID teamID franchID divID Rank  G Ghome  W  L DivWin WCWin LgWin
## 1  2000  AL   ANA     ANA      W    3 162   81 82 80      N    N    N
## 2  2000  NL   ARI     ARI      W    3 162   81 85 77      N    N    N
## 3  2000  NL   ATL     ATL      E    1 162   81 95 67      Y    N    N
## 4  2000  AL   BAL     BAL      E    4 162   81 74 88      N    N    N
## 5  2000  AL   BOS     BOS      E    2 162   81 85 77      N    N    N
## 6  2000  AL   CHA     CHW      C    1 162   81 95 67      Y    N    N
## 7  2000  NL   CHN     CHC      C    6 162   81 65 97      N    N    N
## 8  2000  NL   CIN     CIN      C    2 163   82 85 77      N    N    N
## 9  2000  AL   CLE     CLE      C    2 162   81 90 72      N    N    N
## 10 2000  NL   COL     COL      W    4 162   81 82 80      N    N    N
##   WSWin  R  AB  H X2B X3B  HR  BB  SO  SB CS HBP SF  RA  ER  ERA  CG  SHO
## 1      N 864 5628 1574 309 34 236 608 1024 93 52 47 43 869 805 5.00 5 3
## 2      N 792 5527 1466 282 44 179 535 975 97 44 59 58 754 698 4.35 16 8
## 3      N 810 5489 1490 274 26 179 595 1010 148 56 59 45 714 648 4.05 13 9
## 4      N 794 5549 1508 310 22 184 558 900 126 65 49 54 913 855 5.37 14 6
## 5      N 792 5630 1503 316 32 167 611 1019 43 30 42 48 745 683 4.23 7 12
## 6      N 978 5646 1615 325 33 216 591 960 119 42 53 61 839 751 4.66 5 7
## 7      N 764 5577 1426 272 23 183 632 1120 93 37 54 45 904 849 5.25 10 5
## 8      N 825 5635 1545 302 36 200 559 995 100 38 64 58 765 700 4.33 8 7
## 9      N 950 5683 1639 310 30 221 685 1057 113 34 51 52 816 775 4.84 6 5
## 10     N 968 5660 1664 320 53 161 601 907 131 61 42 75 897 835 5.26 7 2
##   SV IPouts  HA HRA BBA  SOA  E  DP  FP      name
## 1  46  4344 1534 228 662 846 134 182 0.978  Anaheim Angels
## 2  38  4331 1441 190 500 1220 107 138 0.982  Arizona Diamondbacks
## 3  53  4321 1428 165 484 1093 129 138 0.979  Atlanta Braves
## 4  33  4300 1547 202 665 1017 116 151 0.981  Baltimore Orioles
## 5  46  4358 1433 173 498 1121 109 120 0.982  Boston Red Sox
## 6  43  4351 1509 195 614 1037 133 190 0.978  Chicago White Sox
## 7  39  4364 1505 231 658 1143 100 139 0.983  Chicago Cubs
## 8  42  4369 1446 190 659 1015 111 156 0.982  Cincinnati Reds
## 9  34  4327 1511 173 666 1213 72 147 0.988  Cleveland Indians
## 10 33  4290 1568 221 588 1001 94 176 0.985  Colorado Rockies
##   park attendance BPF PPF teamIDBR teamIDlahman45
## 1 Edison International Field 2066982 102 103 ANA ANA
## 2 Bank One Ballpark 2942251 105 103 ARI ARI
## 3 Turner Field 3234304 101 99 ATL ATL
## 4 Oriole Park at Camden Yards 3297031 95 96 BAL BAL
```

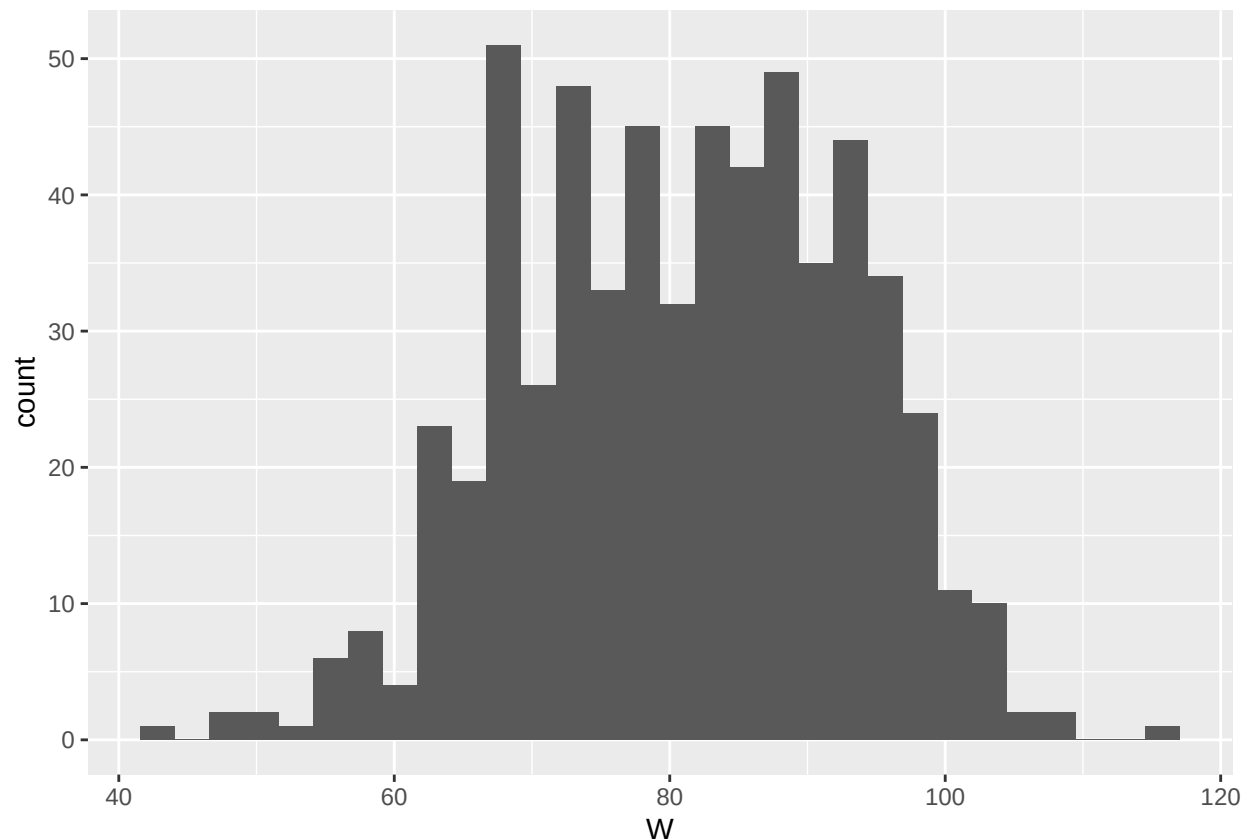
```
## 5          Fenway Park II    2585895 104 103      BOS      BOS
## 6      Comiskey Park II    1947799 102 102      CHW      CHA
## 7          Wrigley Field    2789511  97  98      CHC      CHN
## 8      Cinergy Field      2577371 102 102      CIN      CIN
## 9          Jacobs Field    3456278 101 100      CLE      CLE
## 10         Coors Field     3295129 125 125      COL      COL
##   teamIDretro
## 1          ANA
## 2          ARI
## 3          ATL
## 4          BAL
## 5          BOS
## 6          CHA
## 7          CHN
## 8          CIN
## 9          CLE
## 10         COL
```

The dimensions of this filtered data frame is 600 x 48.

10. Plot the distribution of wins. Describe the shape.

```
Teams %>%
  filter (yearID >= "2000", G>60) %>%
  ggplot(aes(x = W)) + geom_histogram()
```

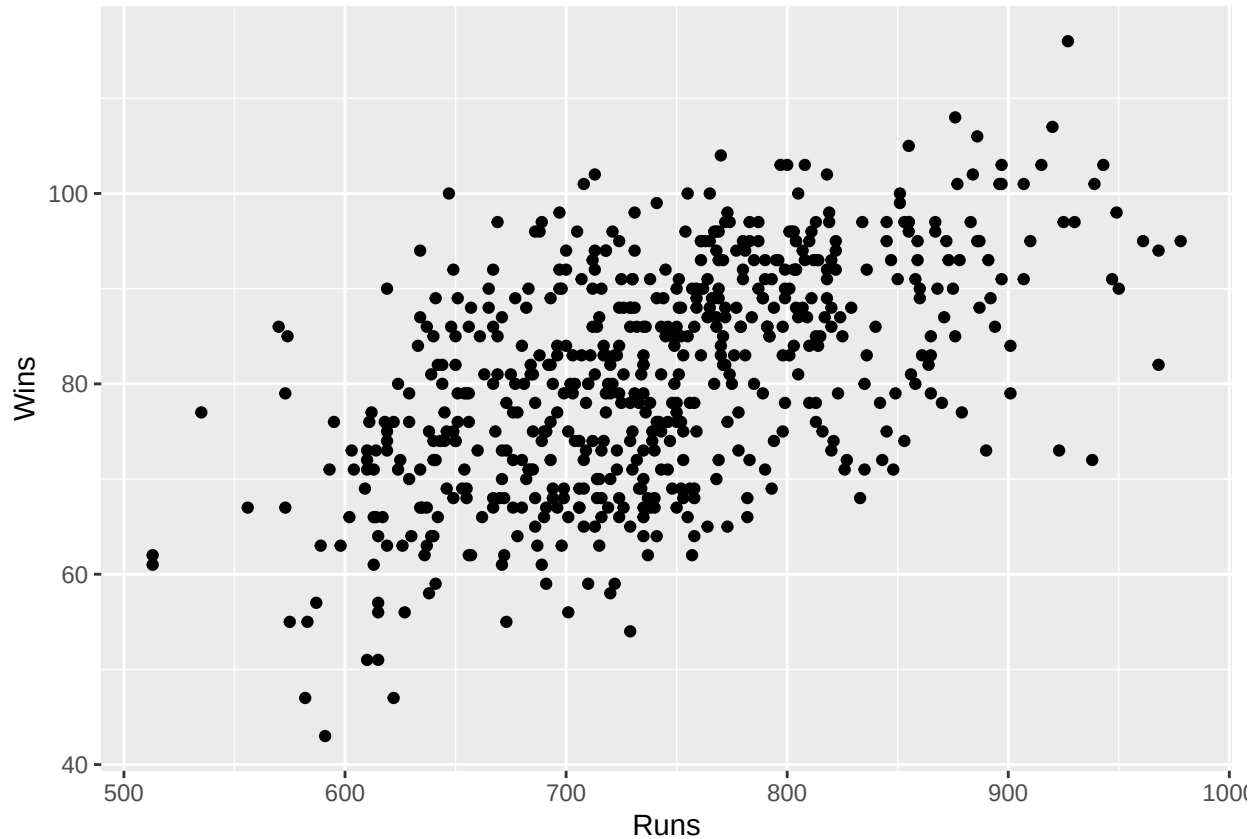
```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```



The histogram shape looks bell-shaped, symmetric, and multimodal, not very skewed.

11. Plot the relationship between runs and wins. Describe the relationship (form, direction, strength, presence of outliers).

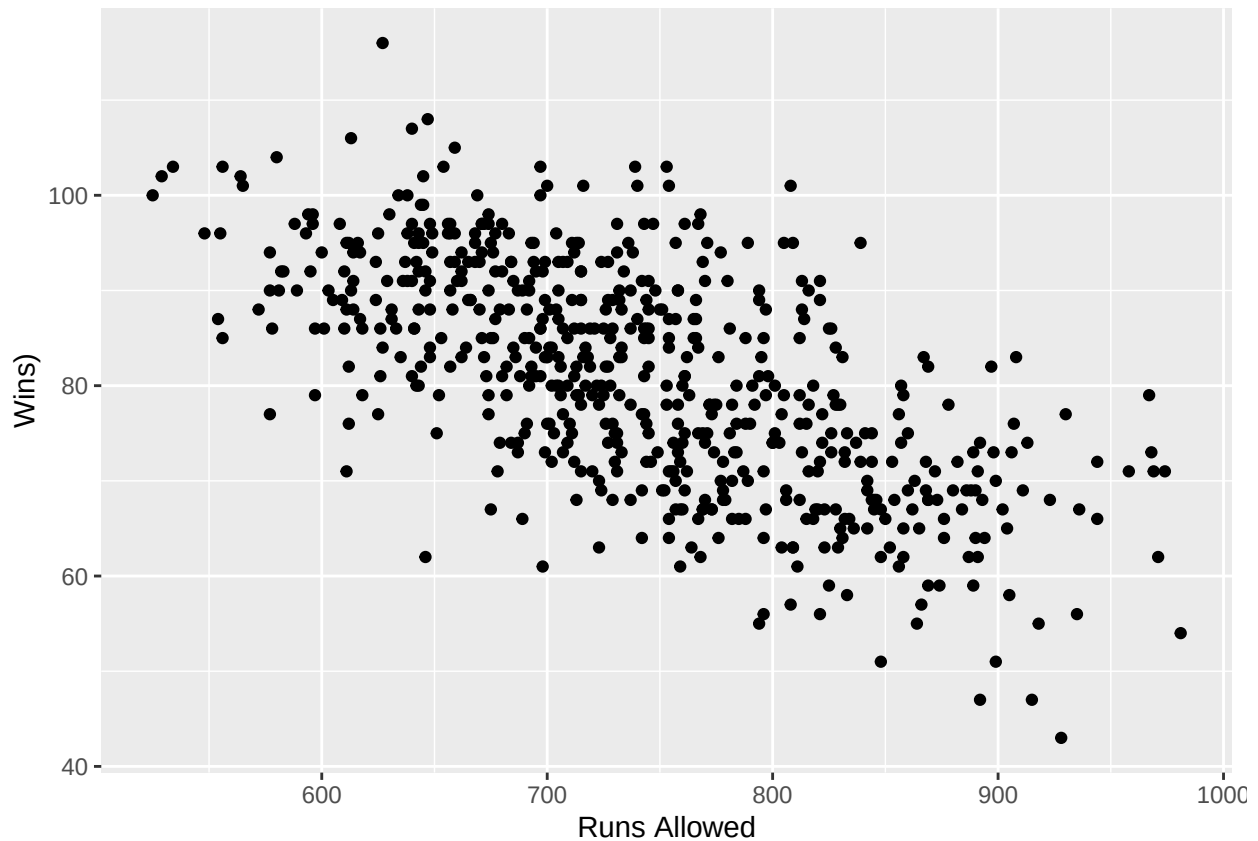
```
Teams %>%  
  filter (yearID >= "2000", G>60) %>%  
  ggplot (aes (x = R, y = W)) + geom_point() + labs(y= "Wins", x = "Runs")
```



The relationship between Runs and Wins seems to be strong, positive and linear. It does not seem to have any conspicuous presence of outliers.

12. Plot the relationship between runs allowed and wins. Describe the relationship. How does it compare to the relationship between runs and wins?

```
Teams %>%  
  filter (yearID >= "2000", G>60) %>%  
  ggplot (aes (x = RA, y = W)) + geom_point() + labs(y= "Wins)", x = "Runs Allowed")
```

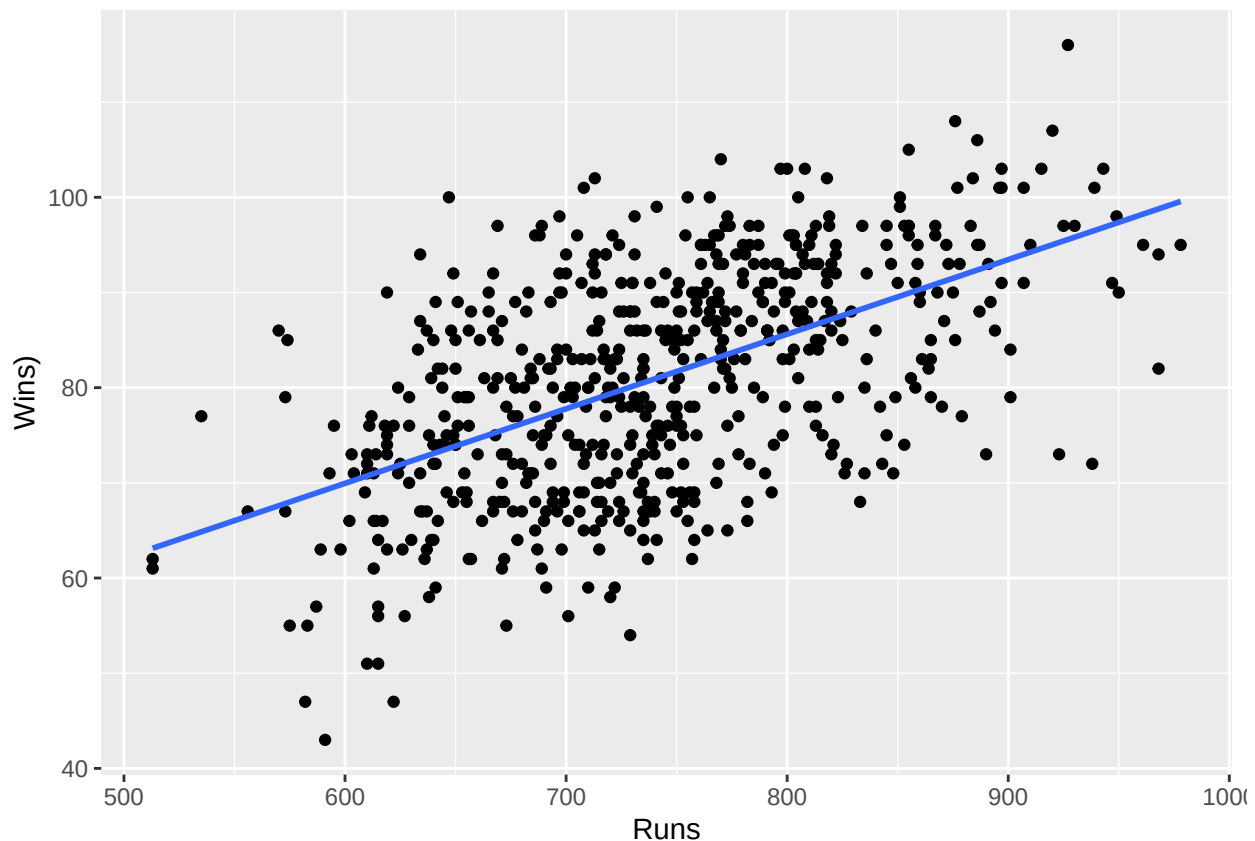


The relationship between Runs Allowed and Wins seems to be strong, negative, and linear. It does not seem to have any conspicuous presence of outlier. Both the relationship between Runs and Wins, and Runs Allowed and Wins are strongly correlated, linear, with no conspicuous presence of outlier. However, the direction is different: relationship of Runs and Wins is positive, but the relationship of Runs Allowed and Wins is negative.

13. Fit a simple linear model to predict wins by runs. Write out the equation for the linear model and interpret the slope. What are the R² and adjusted-R² values?

```
Teams %>%
  filter (yearID >= "2000", G>60) %>%
  ggplot (aes (x = R, y = W)) + geom_point() + labs(y= "Wins)", x = "Runs") + stat_smooth(method = "lm")

## 'geom_smooth()' using formula 'y ~ x'
```



```
teamfilter<-Teams %>%
  filter (yearID >= "2000", G>60)

m1 <- lm( W ~ R, data = teamfilter)
coefficients(m1)
```

```
## (Intercept)      R
## 22.93229805  0.07835856
```

```
summary(m1)
```

```
##
## Call:
## lm(formula = W ~ R, data = teamfilter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -26.2422  -7.2721   0.8249   6.7876  26.3697
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 22.932298   3.596320   6.377 3.62e-10 ***
## R           0.078359   0.004825  16.239 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.827 on 598 degrees of freedom
## Multiple R-squared:  0.306, Adjusted R-squared:  0.3049
```

```
## F-statistic: 263.7 on 1 and 598 DF, p-value: < 2.2e-16
```

The equation of the linear model: $\text{Wins} = 22.93 + 0.078 \times \text{Runs}$; The intercept = 22.932; The R-square value is 0.306. The adjusted R-square value is 0.3049.

14. What is the average number of season runs and wins? Based on the previous model, how many games would you predict a team that scored the average number of runs would win? What about a team that scored 600 runs? What about 850 runs?

```
teamfilter%>%
  summarize(mean_runs = mean(R), mean_wins = mean(W))
```

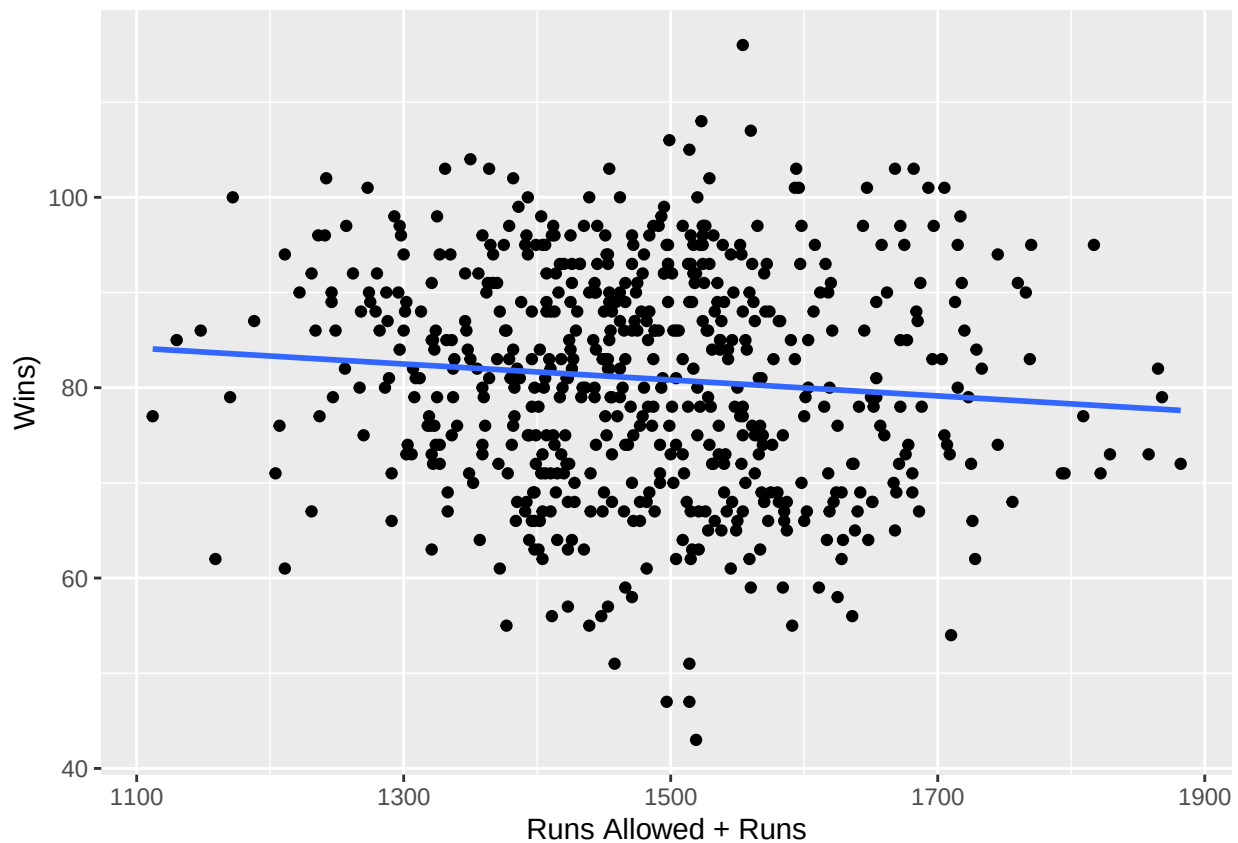
```
##   mean_runs mean_wins
## 1   740.6683    80.97
```

The average number of season runs is 740.6683, the average number of season wins is 80 (80.97). Based on the previous model, I would expect the team that scored the average number of runs to win 80 (80.70) games. If the team scored 600 runs, they would win 69 (69.73) games. If the team scored 850 runs, they would win 89 (89.23) games.

15. Fit a simple linear regression model to predict wins by runs and runs allowed. Write out the equation for the linear model and interpret the slopes. What are the R² and adjusted-R² values? Compare to The simple linear regression from the previous question

```
Teams %>%
  filter (yearID >= "2000", G>60) %>%
  ggplot (aes (x = R + RA, y = W)) + geom_point() + labs(y= "Wins", x = "Runs Allowed + Runs") + stat_

## 'geom_smooth()' using formula 'y ~ x'
```



```
teamfilter<-Teams %>%
  filter (yearID >= "2000", G>60)

m2 <- lm( W ~ R + RA, data = teamfilter)
coefficients(m2)
```

```
## (Intercept)          R          RA
## 85.00923844  0.09671988 -0.10217339
```

```
summary(m2)
```

```
##
## Call:
## lm(formula = W ~ R + RA, data = teamfilter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.2559  -2.8062  -0.2462   2.6268  13.3453
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  85.009238   1.863952   45.61  <2e-16 ***
## R             0.096720   0.002007   48.20  <2e-16 ***
## RA          -0.102173   0.001877  -54.42  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.028 on 597 degrees of freedom
```

```
## Multiple R-squared:  0.8836, Adjusted R-squared:  0.8832
## F-statistic: 2266 on 2 and 597 DF,  p-value: < 2.2e-16
```

The equation of the linear model: $\text{Wins} = 85.0092 + 0.097 * \text{Runs} - 0.102 * \text{Runs Allowed}$; The intercept is 85.0092; The R-square value is 0.8836. The adjust R-sqre value is 0.8832. Compared to the simple linear regression in the previous question, it is same that both relationship are strongly linearly correlated, but the slope changes for both R and RA; Moreover, this new modal predicts better for wins because it has a larger (closer to 1) R Squared Value and adjusted R Squared Value compared to the R Sqaured value of 0.306, 0.036 and the adjusted R Squared value of 03049

16. Fit a multiple regression model to predict wins using at least two other variables in this data set. How does the R2 and adjusted-R2 values change? Do you think the new model you created predicts wins better?

```
teamfilter<-Teams %>%
  filter (yearID >= "2000", G>60)

m3 <- lm( W ~ BB+SO, data = teamfilter)
coefficients(m3)
```

```
## (Intercept)          BB          SO
## 52.69917603  0.07372205 -0.00877251
```

```
summary(m3)
```

```
##
## Call:
## lm(formula = W ~ BB + SO, data = teamfilter)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -32.717  -7.714   0.138   7.934  26.711
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 52.699176   4.505075  11.698 < 2e-16 ***
## BB           0.073722   0.006113  12.060 < 2e-16 ***
## SO          -0.008773   0.002646  -3.316 0.000969 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.5 on 597 degrees of freedom
## Multiple R-squared:  0.2096, Adjusted R-squared:  0.207
## F-statistic: 79.16 on 2 and 597 DF,  p-value: < 2.2e-16
```

$\text{Wins} = 52.699 + 0.074 * \text{BB} - 0.0088 * \text{SO}$; The intercept is at 52.699; The R-squared value is 0.2096 and the adjusted R-squared value is 0.207. Both R-squared value and adjust R-squared value is smaller (closer to 0) than the previous model, so I believe the new model does not predict wins better.

17. Does this seem like a reasonable setting to conduct statistical inference / make generalizations? To which population might a sabermetrician wish to generalize?

This seems like a reasonable setting to conduct statistical inference and make generalization. However, a sabermetrician must be careful of overgeneralizing. Our prediction model are based on data from professional baseball teams in a certain league, so we can only use out data to make predictions about the same teams

and players that we have data for. A sabermetrician may wish to generalize a tee-ball league, they may find our model inaccurate.

18. Is it reasonable to draw a causal conclusion from these models? Why or why not or under which circumstances?

Theoretically, it is not reasonable to draw a causal conclusion from these models, because, our data is observational, not experimental, meaning that we did not control variables to isolate a cause-effect relationship between the two variables of interest. However, since we are able to predict the correlation, under circumstances which the general rules of baseball are followed (in baseball, points are given on runs, and wins based on points), it is a logical connection to relate the effect (the number of wins) to the cause (number of runs).